Polarisation of light

Nele Wurster and Hannes Winkler

ABSTRACT

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1 Introduction

Light is an electromagnetic wave whose electric field can exhibit a specific orientation, a property known as polarization. The polarization of light is a fundamental aspect of optics and plays an important role in a wide range of scientific and technological applications. Examples include polarized sunglasses that reduce glare, optical microscopy techniques that enhance image contrast, modern telecommunications systems that use polarization for signal transmission, and emerging quantum technologies where polarization states can serve as carriers of information.
The aim of this experiment is to investigate the fundamental properties of polarized light. Different methods for generating polarized light will be explored, and the effects of various optical elements on its polarization state will be examined. Furthermore, techniques for measuring and analyzing polarization will be introduced. Through these investigations, a deeper understanding of the behavior of polarized light and its practical applications in optical systems will be developed.

2 Malus's law

2.1 Theory of Malus's law

Malus's law describes the intensity of polarized light after passing through a polarizer, it's derivation can be found in the appendix 8. It states that the intensity I of the transmitted light is related to the intensity I_0 of the incident light and the angle θ between the light's initial polarization direction and the axis of the polarizer by the equation:

$$I = I_0 \cos^2(\theta) \tag{1}$$

After passing through the polarizer, the light is polarized along the axis of the polarizer. The wave components that are not transmitted are absorbed by the polarizer, resulting in a decrease in intensity.

2.2 Experimental setup for Malus's law

The unpolarized light of an incandescent lamp was first vertically polarized using a polarizer. The polarized light then passed through a second polarizer, called the analyzer, which can be rotated to change the angle θ between the polarization axis of the polarizer and the analyzer. The angle θ was varied from $0°$ to $180°$ in increments of $10°$, and the corresponding intensity values were measured using a photocell.

2.3 Results of Malus's law

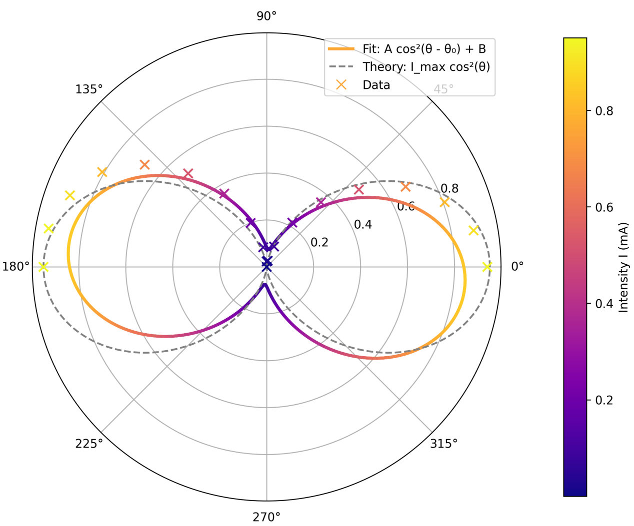

Figure 1: Measured intensity values and the fitted curve with fit $I = A\cos^2(\theta - \theta_0) + B$, $A = 0.7720mA$, $\theta_0 = -5.95°$, $B = 0.0761mA$.

As one can see in fig. 1, for low intensity values, the measured data points fit better to fitted curve, whereas for the angle from $\theta = 0° - 90°$ the data points fit good tho the theoretical curve. This indicates, that the measured intensity values are described sufficiently enough by Malus's law, but there are some systematic errors in the measurement, which can be caused by the fact that the polarizers are not ideal and do not completely absorb the light that is not transmitted. Another source of error could be, that the room was not completely dark, which could have led to additional light being measured by the photocell. Additionally, the angle θ was not measured with high precision, which could have also contributed to the deviations from the theoretical curve.

3 Birefringence of a calcite crystal

3.1 Theory of birefringence

Birefringence, also known as double refraction, is a phenomenon that occurs in certain anisotropic materials, such as calcite crystals. When light enters a birefringent material, it splits into two rays: the ordinary ray and the extraordinary ray. These rays experience different refractive indices due to the anisotropic nature of the material, resulting in different propagation speeds and directions. The ordinary ray follows Snell's law, while the extraordinary ray does not, leading to a separation of the two rays within the crystal.

3.2 Experimental setup for birefringence

In this experiment, a calcite crystal was placed in the path of an unpolarized light beam. As the light passed through the calcite crystal, it split into the ordinary and extraordinary rays. A analyzer was placed behind the crystal to observe the polarization states of the two rays. The rays were projected onto a screen. By rotating the polarizer and observing the changes in intensity and direction of the rays, we can analyze the birefringent properties of the calcite crystal.

3.3 Results of birefringence

4 $\lambda/4$-Plate

4.1 Theory of a $\lambda/4$-plate

A $\lambda/4$-plate, also known as a quarter-wave plate, is an optical device that introduces a phase shift of $\pi/2$ (90 degrees) between the orthogonal components of polarized light. When linearly polarized light passes through a $\lambda/4$-plate, it can be converted into circularly polarized light if the fast axis of the plate is aligned at 45 degrees to the polarization direction of the incident light. Conversely, circularly polarized light can be converted back into linearly polarized light by passing it through a $\lambda/4$-plate with the appropriate orientation.

4.2 Experimental setup for a $\lambda/4$-plate

In this experiment a $\lambda/4$-plate made of quartz was used. The light of an incandescent lamp was first linearly polarized using a polarizer. The linearly polarized light then passed through the $\lambda/4$-plate which was oriented at the angles $\alpha = 0°, -45°, -90°$ with respect to the polarization direction of the incident light. After passing through the $\lambda/4$-plate, the light was analyzed using a analyzer which was rotated to observe the changes in intensity and polarization state of the transmitted light.
Afterwards the $\lambda/4$-plate was oriented at $\alpha = -45°$ and another identical $\lambda/4$-plate was placed in the ray path, such that its optical axis was oriented parallel to the first $\lambda/4$-plate.

4.3 Results of a $\lambda/4$-plate

For $\alpha = 0°$ and $\alpha = -90°$, the transmitted light remained linearly polarized because the light is polarized along the same axis to one of the oplical axes and excites only that component. Thus no relative phase shift occurs and the polarization state does not change. For $\alpha = -45°$, the transmitted light became circularly polarized because the light is polarized at 45 degrees to the optical axes of the $\lambda/4$-plate, exciting both components equally. The $\lambda/4$-plate introduces a phase shift of $\pi/2$ between the two components, resulting in circular polarization. At any other angle $\alpha =]0, -90°[\quad \backslash -45°$ the transmitted light became elliptically polarized because the light is polarized at an angle that is not aligned with the optical axes of the $\lambda/4$-plate, exciting both components but with different amplitudes.
When the second $\lambda/4$-plate was added with its optical axis parallel to the first one, tthe system is equivalent to a $\lambda/2$-plate, which introduces a phase shift of π between the two components. The rotation angle is twice the angle between the incident polarization and the optical axis of the plate. The light remains linearly polarized but with a rotated polarization plane.
If the quartz $\lambda/4$-plate is replaced by a plate made of calcite, the thickness of the plate would need to be adjusted to achieve the same phase shift of $\pi/2$ because the birefringence of calcite is different from that of quartz.

5 Jones-Formalism

The Jones formalism can be used to describe mathematically the polarization state of light and the effect of polarizers and wave plates. Here the polarization state is represented as a two-dimensional vector and the optical elements are represented as a 2x2 matrix.

Linearly polarized light in x-direction: $\vec{J}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Linearly polarized light in y-direction: $\vec{J}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Linearly polarized light at 45°: $\vec{J}_{45} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Right circularly polarized light: $\vec{J}_R = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$

Left circularly polarized light: $\vec{J}_L = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$

The Jones matrix for the $\lambda/4$-plate with the fast axis along the x-direction is given by:

$$\mathbf{M}_{\lambda/4} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1-i & 0 \\ 0 & 1+i \end{pmatrix}$$

This matrix introduces a phase shift of $\pi/2$ between the x and y components of the electric field, which is characteristic of a quarter-wave plate. When linearly polarized light at 45 degrees passes through this $\lambda/4$-plate, it becomes circularly polarized. Conversely, circularly polarized light can be converted back into linearly polarized light by passing it through the same $\lambda/4$-plate swith the appropriate orientation.

6 Circular birefringence of sucrose

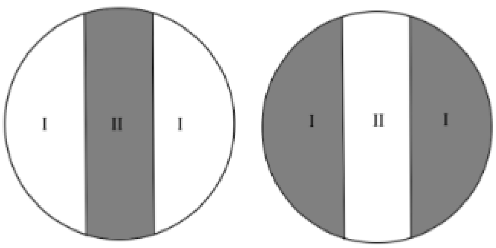

Figure 2: Looking through the polarimeter, there are two fields visible, one at the edges and one in the middle. Field I has a slightly different polarisation angle than field II. The light from both fields passes through an adjustable polarisation filter, causing a difference of brightness.

6.1 Theory

In many fields of study concentration analysis of unknown solutions is fundamental. To quickly determine the concentration of a solution, one can use the method of half-shadow polarimeter. This method is based on the fact that certain substances, such as sucrose, exhibit circular birefringence, which causes the plane of polarization of light to rotate as it passes through the solution. The angle of rotation is directly proportional to the concentration of the solution and the path length of the light through it. By measuring the angle of rotation, one can determine the concentration of the solution.
For that method, the penumbral angle of the used polarimeter has to be determinded, by measuring the position of minimal brightness in field I and II and then subtracting the two angles from each other(see fig. 2).

6.2 Measurements and results

The penumbral angle of the used polarimeter was determined to be $\psi = (11.60 \pm 0.73)°$.
Now the concentration of an unknown sucrose solution had to be determined. For that, we sent light with wavelength $\lambda = 595nm$ through a solution of sucrose in water with concentration $c_{100} = 100g/l$, examined it's optical rotation which resulted $\theta_{100} = (14.00 \pm 0.75)°$ and afterwards determined the rotation of a solution with unknown concentration to $\theta_{unknown} = (9.00 \pm 0.59)°$.
The sucrose concentration of the unknown solution can be calulated using the formula:

$$c_{unknown} = \frac{\theta_{unknown}}{\theta_{100}/c_{100}} \tag{2}$$

With the measured values, the concentration of the sucrose was determined to $c_{unknown} = (64.3 \pm 6.8)\frac{g}{l}$. This seems reasonable for the amount of water, the c_{100} solution was diluted with.

7 Measuring the Faraday effect inside a glass body

8 Appendix

Derivation of Malus's Law

Consider linearly polarized light with intensity I_0 and polarization direction $\vec{e}_P$ incident on an analyzer whose transmission axis $\vec{e}_A$ is rotated by an angle θ relative to $\vec{e}_P$. Only the component of the electric field along $\vec{e}_A$ is transmitted.

The incident electric field can be written as

$$\vec{E} = E_0\, \vec{e}_P, \tag{3}$$

where E_0 is the amplitude. The transmitted component along $\vec{e}_A$ is

$$E_t = \vec{E} \cdot \vec{e}_A = E_0 \cos\theta. \tag{4}$$

Since the intensity is proportional to the square of the field amplitude, $I \propto E^2$, it follows that

$$I(\theta) = I_0 \cos^2\theta \tag{5}$$

Mean Intensity after the Analyzer

If unpolarized light of intensity I_{in} is incident on the polarizer, an ideal polarizer transmits exactly half of it:

$$I_0 = \frac{I_{\text{in}}}{2}. \tag{6}$$

This follows from averaging $\cos^2\theta$ over all polarization directions:

$$\langle \cos^2\theta \rangle = \frac{1}{2\pi}\int_0^{2\pi} \cos^2\theta \,\mathrm{d}\theta = \frac{1}{2\pi}\int_0^{2\pi} \frac{1+\cos 2\theta}{2}\,\mathrm{d}\theta = \frac{1}{2}. \tag{7}$$

Therefore, the intensity of the light incident on the analyzer is

$$I_0 = \frac{I_{\text{in}}}{2} \tag{8}$$

i.e. the intensity after the polarizer is exactly half of the incoming intensity, independent of the orientation of the analyzer.

References

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